



Intel Core i9-11900 Specs

A Bilibili user has leaked specifications for three 11th Gen Intel Rocket Lake-S processor samples. Find out more! <http://digit.in/jan21-73>



Super Nintendo Portable

Using an SNES controller as housing, and Raspberry Pi to power the display, a fan creates a portable console. <http://digit.in/jan21-74>

One too many

Agent001 troubleshoots a PC for a first timer

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he whole work from home situation has caused many of my friends to finally wake up to see the

limitations of their existing machines. And it's not just their laptops but also their networking gear and everything else that most folks take for granted have all shown their true colours. Processing power of laptops has improved by leaps and bounds but the recent tax hikes has changed the performance-per-rupee factor by a significant amount. And when you're talking about networking gear, people have realised that spending less on a cheap router isn't exactly worth it since a lot more devices are now simultaneously connected to one router and the cheap ones are often artificially gated to help sell the better units. So it's natural for devices to mysteriously lose connection or for the router to not properly prioritise the right connections.

Thankfully, folks have started procuring the good stuff. By that, I don't mean the expensive gear but hardware that is more appropriate for their needs. One such friend ended up building his first PC after watching a couple of YouTube videos and reading up a bit. It's great to see more folks get in on the DIY scene. What surprised me was that my friend termed his fresh PC to be "not

that big an improvement" over his existing laptop. This had me quite bewildered since his PC had the latest 10th Gen Intel processor, 16 GB of RAM, an RTX 2060 Super and an NVMe SSD with over 3000 Mbps of transfer speed. Calling that PC to be "not that big an improvement" over a four year-old laptop was a little puzzling. So I had to hop over to his place to see exactly what was wrong.



PC building is quite complicated for first time builders even though things have become immensely simpler over time. Most videos and tutorials available online don't cover the nuances of PC building and it's easy for builders to commit mistakes unless folks do a lot of research. As I started going through the operating system, it seemed like my friend had installed the OS properly and all drivers were installed. The only thing is everything was a little out of place. We ran a few benchmarks and

scores were quite lower than what the hardware was capable of. For starters, the RAM modules were running at their lowest spec since XMP (Extreme Memory Profile) was not switched on. Modules rated for 3600 MHz were installed in the wrong slots and were not operating in dual-channel mode. This was an easy fix and we ran a few benchmarks and saw a decent 8-11 per cent improvement.

As I was navigating through the BIOS, I saw a few other things that seemed out of place. The hard drives had been connected to SATA ports that shared bandwidth with the M.2 port. And that M.2 port was populated with the only SSD. So while the hard

drives were functioning properly, the NVMe SSD was operating with just 2x lanes. So transfer speeds were really low. The M.2 SSD was moved to the primary port which did not share bandwidth and we manually set the port to PCIe Gen 3 from the default Auto position. We then saw transfer speeds as advertised by the SSD manufacturer. The graphics card was installed in the proper slot and functioning normally so it seemed the RAM misconfiguration was the primary culprit.